$f(6) \le 24$: an unconditional in-house upper bound for almost-equidistant sets in $\mathbb{R}^6$

The frontier argument over the independently verified ledger

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Statement

A finite point set in $\mathbb{R}^d$ is *almost equidistant* if among any three of its points two are at unit distance; $f(d)$ is the maximum size. The published bounds for $d = 6$ are $18 \le f(6) \le 26$ [1]. The companion note `f6_le_20.pdf` (same date) proves $f(6) \le 20$ *conditional* on a certificate ledger that includes exact-filter layers of the `codex/dimension6` track not yet independently reimplemented. This note records the stronger-trust, weaker-value counterpart obtained the same day:

Result. $f(6) \le 24$*, on an entirely in-house trust base: (i) the verified* 3,971,787*-graph level-19 candidate corpus; (ii) the first-generation exact graph filters, independently reimplemented from their theory document with exact agreement on all per-rule corpus statistics, and with hand-checked derivations; (iii) this repository's* 3,055,480 *interval-arithmetic non-realizability certificates (certified engine* `ckernel6.c`*); (iv) the extension/frontier pipeline, double-implemented (C and Python) with exact agreement on every stage of every run, and validated against two independent corpus controls. Computer-assisted; not peer-reviewed.*

The computation

Let $\mathcal{R}_{\mathrm{ih}}$ be the 114,242 level-19 candidates not rejected by (ii) or (iii) — the in-house unresolved residue (`residue_inhouse.txt`; it contains the 644-graph residue of the conditional note as a subset). By the extension theorem and hereditary frontier lemma of `f6_le_20.pdf`, every n-point almost-equidistant set ($n \ge 20$) induces a member of the level-n *frontier* over $\mathcal{R}_{\mathrm{ih}}$: the iteratively defined set of level-n candidates all of whose single-vertex deletions re-complete (in every maximal triangle-free completion) into the previous level's frontier. The computation (tools `ext6.c` modes `extend` / `frontier` / `escalate`; $\sim$25 minutes wall time) gives:

level	distinct extensions	greedy survivors	true frontier
20	6,419,680	2,377	2,356
21	142,963	127	127
22	3,028	23	23
23	82	2	2
24	2	2	2
25	0	—	$\emptyset$

The two level-24 frontier graphs admit no valid one-vertex extension at all, so the level-25 frontier is empty: no 25-point almost-equidistant set exists in $\mathbb{R}^6$, and by heredity $f(6) \le 24$.

The tower, and the ladder ahead

The surviving tower is remarkable: the level-23 pair has 176 edges against $6 \cdot 23 - 21 = 117$ degrees of freedom, the level-24 pair 192 against 123; their complements are triangle-free graphs with independence number at most seven, close to the $R(3,8)=28$ critical regime. A Levenberg–Marquardt screen (800 restarts) puts all four far from realizability (best squared residuals 2.12–3.36; realizable controls reach 10^{-30}).

Each tower level is an explicit in-house certification target: proving the 2 level-23 graphs non-realizable gives $f(6) \le 22$; the 23 level-22 graphs give $f(6) \le 21$; the 127 level-21 graphs give $f(6) \le 20$, matching the conditional bound without its conditional tier. (The interval kernel currently compiles with MAXN $= 20$ and must be rebuilt for larger n before those runs.)

Reproduction

```
clang -O2 -o ext6 ext6.c
./indep_profile6 aeq_d6_n19.txt v.bin        # exact-filter verdicts
python - <<'PY'                              # residue_inhouse.txt
# (see extract in repository history; verdict bit7 clear and not in
#  killed_d6_n19.log)
PY
./ext6 extend residue_inhouse.txt - > ext20_ih_raw.txt
bash run_ih_frontier.sh 6                    # levels 20-21
# then iterate extend/frontier/escalate to levels 22-25 as in STATUS.md
```

References

[1] M. Balko, A. Pór, M. Scheucher, K. Swanepoel, P. Valtr, Almost-equidistant sets, *Graphs and Combinatorics* 36 (2020), 729–754.